



The AI Explosion: Can Artificial Intelligence Trigger Unprecedented Economic Growth?

【经济学人播客 AI引爆经济革命：从蒸汽时代到智能时代 - 增长奇迹还是泡沫幻影？
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Summary: The podcast explores the potential for AI to trigger explosive economic growth, examining historical precedents like the Industrial Revolution. Experts discuss Silicon Valley's predictions of 20-30% annual growth rates through AI replacing human labor. They analyze economic models showing how AI worker accumulation could surpass human population growth constraints. The discussion covers potential bottlenecks, workforce disruption, and investment implications. While acknowledging AI's transformative potential, they caution that real-world constraints may prevent the most extreme growth scenarios.

摘要：本期播客探讨了AI引爆爆炸性经济增长的潜力，并回顾了工业革命等历史先例。专家们讨论了硅谷对AI取代人力实现20-30%年增长率的预测，分析了AI劳动者积累如何突破人口增长限制的经济模型。对话涉及发展瓶颈、劳动力市场冲击和投资影响等关键问题。虽然承认AI的变革潜力，但专家指出现实约束可能阻止最极端的增长情景。





Can AI supercharge global economic growth?



🕒 Estimated Reading Time: 19 min.

📖 四级生词 📖 六级生词 📖 雅思生词 📖 托福生词 📖 专八生词 📖 SAT生词 📖 考研生词 📖 GRE生词 📖 其它生词

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🎵 In 1700, it would have seemed natural that economies only ever stood still.

在1700年，人们很自然地认为经济只会停滞不前。

🎵 In the common era up to that point, output had expanded by just 0.1% per year on average.

在那之前的公元时代，产出平均每年仅增长0.1%。

🎵 Then, steam engines began to puff.

随后，蒸汽机开始喷涌蒸汽。

▶ Global growth quintupled to 0.5% per year between 1700 and 1820.

在1700年至1820年间，全球增长率翻了五倍，达到每年0.5%。

▶ By the end of that century, it had reached 1.9%.

到那个世纪末，增长率达到了1.9%。

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▶ And in the 20th century, output grew by 2.8% on average.

而在20世纪，产出的平均增长率为2.8%。

▶ The long history of the world economy is one of expansion at an increasing rate.

世界经济漫长的历史是一部增速不断加快的扩张史。

▶ And if you believe the inhabitants of Silicon Valley, the world's economy is about to be transformed again.

而如果你相信硅谷居民的说法，世界经济即将再次发生变革。

▶ And that's because AI is going to get so powerful that it is going to be able to carry out all sorts of tasks that are currently done by humans and eventually all tasks.

这是因为人工智能将变得如此强大，以至于它将能够执行目前由人类完成的各种任务，并最终完成所有任务。

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▶ And the implication of that is that you get explosive economic growth.

而这其中的含义是，你将获得爆炸性的经济增长。

▶ I'm Jason Palmer, co-host of The Intelligence Podcast.

我是杰森·帕尔默，《情报》播客的联合主持人。

▶ And I'm Henry Kerr, economics editor, The Economist.

我是亨利·克尔，《经济学人》的经济学编辑。

▶ Today, we're going to carry out a little thought experiment looking into how the world economy will develop if the most outlandish predictions of the AI world actually come true.

今天，我们将进行一个小小的思维实验，探讨如果人工智能领域最离谱的预测成真，世界经济将如何发展。

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▶ Remember to listen to the full episode.

记得收听完整节目。

▶ Click the link in the description below.

点击下方描述中的链接。

▶ Henry, let's just start with what the predictions are as they stand now.

亨利，我们先从目前的预测说起。

▶ Well, if you believe the inhabitants of Silicon Valley, the world economy is about to be transformed.

嗯，如果你相信硅谷居民的说法，世界经济即将发生变革。

▶ And that's because AI is going to get so powerful that it is going to be able to carry out all sorts of tasks uh that are currently carried out by humans and eventually all tasks that are currently carried out by humans.

这是因为人工智能将变得如此强大，以至于它将能够执行目前由人类完成的各种任务，并最终完成所有目前由人类完成的任务。

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▶ And the implication of that in some economic models is that you get completely explosive economic growth.

在一些经济模型中，这其中的含义是，你将获得完全爆炸性的经济增长。

▶ We're used to growth of 2% or maybe 3% in a good year in advanced economies over the past half century or so.

在过去半个世纪左右的时间里，我们已经习惯了发达经济体2%或好年景时3%的增长。

▶ Uh some of these models churn out numbers like 20 or 30% growth.

呃，其中一些模型给出的数字是20%或30%的增长。

▶ Carrying out our thought experiment then how do we get from a 2% 3% world to a 20 30% world?

那么进行我们的思维实验，我们如何从一个2%-3%增长的世界过渡到一个20%-30%增长的世界？

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▶ What does the middle of that journey look like?

这个过程的中阶段是怎样的？

▶ Well, I think the best way to answer that is to think about what enabled humanity to go from the very low growth pre-industrial revolution to the two to three% growth norm today.

嗯，我认为回答这个问题的最好方法是思考是什么让人类能够从工业革命前极低的增长过渡到今天2%到3%的增长常态。

▶ And part of that picture early on was the so-called accumulation of labor.

早期情况的一部分是所谓的劳动力积累。

▶ Uh the size of the economy was very closely linked to the size of the population.

呃，经济规模与人口规模密切相关。

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▶ And a bigger population led to the creation of more ideas because you have more people sitting around thinking uh and more output enabled death rates to come down and enabled birth rates early in the the industrial revolution.

更大的人口规模催生了更多的想法，因为有更多的人在思考，呃，而更多的产出使得死亡率得以降低，并在工业革命早期使得出生率得以提高。

▶ In theory, if your labor force, as you might call it, is made up of lots of AI agents who are really capable.

理论上，如果你所谓的劳动力是由许多真正有能力的人工智能体组成的。

▶ In theory, you don't have to wait around for generations to pass for your workforce to grow in the same way as you would have done in the earlier era.

理论上，你不需要像早期时代那样等待几代人的时间让劳动力增长。

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▶ So the people who come up with these kind of numbers talk about the accumulation of AI workers being far more rapid than the accumulation of human workers.

所以，提出这类数字的人谈论的是人工智能工人的积累速度远快于人类工人的积累速度。

▶ So what starts to happen is you get massive investment in uh the the production of AI agents, data centers, energy and so on.

于是开始发生的情况是，你会看到对人工智能体生产、数据中心、能源等的大规模投资。

▶ That pays off and produces a lot of growth because it automates so many tasks and then that payoff is reinvested into still more AI power and that loop can turn really quickly unlike when you're looking at population accumulation uh where it takes a while for that to happen.

这会带来回报并产生大量增长，因为它自动化了许多任务，然后这些回报又被再投资到更多的人工智能能力上，这个循环可以转得非常快，不像人口积累那样需要一段时间才能发生，呃。

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▶ So two things there.

所以这里有两件事。

▶ One is that for the people who work um in automatable tasks, they are then going to have trouble more quickly.

一是对于那些从事可自动化任务的人来说，他们很快就会遇到麻烦。

▶ But that still leaves a bunch of things that AIs can't do.

但这仍然留下了一堆人工智能还做不到的事情。

▶ AI plumbers don't yet exist.

人工智能水管工还不存在。

▶ Yes.

是的。

▶ So there are various extremes you to which you can push this thought experiment.

所以你可以把这个思维实验推向各种极端。

▶ Lots of people in Silicon Valley believe this.

硅谷的很多人都相信这一点。

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▶ Super intelligence basically replaces everybody.

超级智能基本上会取代所有人。

▶ Uh you have an AI that makes itself better.

呃，你有一个能自我改进的人工智能。

▶ uh you have an AI that solves all robotics engineering challenges and you don't need humans anymore.

呃，你有一个能解决所有机器人工程挑战的人工智能，你就不再需要人类了。

▶ In reality, what you're much more likely to see is intermediate phases which last a very long time in which there are bottlenecks.

在现实中，你更可能看到的是持续很长时间的中间阶段，其中存在瓶颈。

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▶ That might be bottlenecks to how fast the AIs can improve or how much investment in AI can take place.

这些瓶颈可能是人工智能的改进速度能有多快，或者能进行多少人工智能投资。

▶ Or it may simply be um uh things like regulatory bottlenecks to having robots displace uh humans.

或者可能仅仅是，嗯，呃，诸如监管瓶颈之类的东西，阻碍机器人取代，呃，人类。

⏮ uh or it might be that there are fundamental limits to what AIs and and robots can do.

呃，或者可能是人工智能和机器人能力存在根本性的限制。

⏮ And uh typically what's happened in the history of the world economy is that when you have very rapid productivity growth in one type of activity and not much productivity growth in another type of activity, then the low productivity growth sectors still see wages go up a lot.

而呃，世界经济史上通常发生的情况是，当某一类活动的生产率增长非常快，而另一类活动的生产率增长不大时，那么低生产率增长的部门仍然会看到工资大幅上涨。

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⏮ Uh and this is referred to as as cost disease.

呃，这被称为成本病。

⏮ Uh but if you're a worker being displaced by an AI and you're having to find a new job, you might quite like cost disease because it means there's lots of highly remunerated stuff still around for you to do.

呃，但如果你是一个被人工智能取代、不得不找新工作的工人，你可能会相当喜欢成本病，因为这意味着周围仍然有很多高报酬的工作可供你去做。

⏮ But uh that's still a lot of disruption happening if you have workers moving from one sector to another as uh AIS expand and and displace more uh of the labor force.

但是呃，如果随着人工智能的扩展并取代更多的劳动力，工人从一个部门转移到另一个部门，这仍然会造成很大的混乱。

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⏮ So looking at um Silicon Valley valuations on these AI companies and so on.

那么，看看呃硅谷对这些人工智能公司等估值。

⏮ Should I in your view become a plumber or buy into these companies?

依你之见，我是应该去当水管工，还是应该投资这些公司？

⏮ Well, I wouldn't become a plumber just yet on the basis of a thought experiment.

嗯，我不会仅仅基于一个思维实验就立刻去当水管工。

▶ You could buy into the companies.

你可以投资这些公司。

▶ There are a lot of people in Silicon Valley who think that's essential because in the long run they think labor is not going to have any value.

硅谷有很多人认为这很必要，因为他们认为从长远来看，劳动力将不再有任何价值。

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▶ The only thing that's going to have any value is uh owning capital, having a stake in these AI companies and things that are complimentary to them.

唯一会有价值的东西是呢拥有资本，持有这些人工智能公司及其互补品的股份。

▶ Um and that is uh justified by what you find in some of the economic models of explosive growth based on automation.

嗯，这一点呢可以从一些基于自动化的爆炸性增长经济模型中得到印证。

▶ The returns to capital go up.

资本回报率上升。

▶ But it's a little bit complicated by the fact that those economic models also show uh that in an explosive growth scenario, interest rates should go up a lot.

但有点复杂的是，那些经济模型也显示呢，在爆炸性增长的情景下，利率应该会大幅上升。

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▶ And one way of thinking about this is is if you're going to have explosive economic growth, you're and you're going to have AI powering that, you need a ton of data centers.

思考这个问题的一种方式，如果你要实现爆炸性的经济增长，并且由人工智能驱动，你就需要大量的数据中心。

▶ you need a lot of energy production.

你需要大量的能源生产。

▶ Uh if you're going to have a big economy, you need more infrastructure.

呃，如果你要有一个庞大的经济，你就需要更多的基础设施。

▶ Uh so there's all sorts of demands for capital.

呃，所以会有各种各样的资本需求。

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▶ But you have a lot of people who think they're about to get rich because of AI sending economic growth to the moon.

但你有很多人认为他们即将因为人工智能推动经济增长至月球而致富。

▶ And so no one really wants to save because why save for tomorrow if you're going to be rich tomorrow?

所以没有人真的想储蓄，因为如果你明天就会变富，为什么还要为明天储蓄？

▶ A reduced demand to save increased demand for investment means there's a kind of capital shortage that pushes up interest rates.

储蓄需求减少，投资需求增加，意味着一种资本短缺，从而推高利率。

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▶ And what do higher interest rates tend to do?

而更高的利率往往会做什么？

▶ They tend to reduce asset prices.

它们往往会降低资产价格。

▶ So it's a it's it's you know you can tell the story multiple ways here and uh it's it's not entirely clear even if you got capital exactly what you should buy if what you think is going to happen is explosive growth because it depends on these obscure parameters in the models which economists and finance professors disagree about coming out of the models and looking at the real world what what to watch.

所以这是一个，你知道，你可以用多种方式来讲述这个故事，而且呃，即使你拥有资本，如果你认为将要发生的是爆炸性增长，也不完全清楚你到底应该买什么，因为这取决于模型中那些晦涩

的参数，经济学家和金融教授对这些参数意见不一，从模型出发，观察现实世界，应该关注什么。

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▶ How how do you sort of figure out which which end of things are we're going to actually see as some of this stuff does or maybe doesn't come to fruition?

你如何大致判断，随着其中一些事情实现或可能未实现，我们实际上会看到哪种结果？

▶ I think the interest rate story is quite important here because although it is the case at the moment that you have this very uh extreme abalance in the US stock market, high valuations of AI firms which would you know seem to suggest the market buying into the thesis somewhat.

我认为利率的故事在这里相当重要，因为尽管目前美国股市确实存在这种非常呃极端的不平衡，人工智能公司估值很高，这你知道似乎暗示市场在某种程度上认同这个论点。

▶ in the uh money markets you do not have the pricing in of explosive economic growth and a big increase in interest rates and and I think that interest rate story is quite compelling in theory.

但在呃货币市场上，你并没有看到对爆炸性经济增长和利率大幅上升的定价，而且我认为利率的故事在理论上是相当有说服力的。

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▶ So the thing you should watch for whether or not markets are really starting to believe an explosive growth story would be the uh long-term bond yield which should should rise quite a lot if uh if explosive growth is coming.

所以，你应该关注的是市场是否真的开始相信爆炸性增长的故事，这要看呃长期债券收益率，如果呃如果爆炸性增长即将到来，它应该会大幅上升。

▶ If it's just really valuable AI companies, well, you can explain that, I think, with a story where yes, these companies are really profitable and AI is the next big big thing, but the economy as a whole isn't exploding.

如果只是非常有价值的人工智能公司，嗯，我想你可以用一个故事来解释，即是，这些公司确实非常赚钱，人工智能是下一个重大事件，但整个经济并没有爆炸式增长。

▶ It's just that AI is the latest technology that keeps the economy on a on a normal growth path.

只是人工智能是最新的技术，使经济保持在正常的增长轨道上。

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▶ It happens to be AI today, uh, just like in the past it would have been manufacturing, automation, uh, or the internet or or or or whatever, electricity.

今天碰巧是人工智能，呃，就像过去可能是制造业、自动化，呃，或互联网，或随便什么，电力。

▶ Um, and today it's going to be AI.

嗯，而今天将是人工智能。

▶ watch the bond markets.

关注债券市场。

▶ I'd say at the mention of the internet that was there was as much froth around this the the notion of the internet when it was new and how it was going to change productivity, how it's going to change the world, change the markets, change the economies and so on and kind of didn't in the long run.

我得说，提到互联网，当互联网还是新事物时，围绕它的概念——它将如何改变生产力、如何改变世界、改变市场、改变经济等等——也存在过同样多的泡沫，但从长远来看，它并没有做到。

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▶ Is that instructive to your mind or is AI a sufficiently different technology that we should consider it differently?

这对你来说是否有指导意义，还是说人工智能是一种足够不同的技术，我们应该以不同的方式看待它？

▶ Yes.

是的。

🎧 Well, I found myself hesitating when I mentioned the internet there for precisely uh precisely this reason.

嗯，我刚才提到互联网时发现自己犹豫了，正是呢正是因为这个原因。

🎧 Um it it It's clear that the internet both transformed the world and did not impact economic growth all that much at least in a way that's observable and some people do tell this story where AI will be a similar sort of thing lots of disruption not much measured economic growth and the economists who studied this in the 2010s uh tended to come up with estimates where essentially all the benefit of the internet and uh social media free services went to consumers but uh uh but but not in a way that shows up in your measured economic output.

嗯，很明显，互联网既改变了世界，又对经济增长没有产生太大影响，至少以可观察的方式是这样，有些人确实在讲述这样一个故事：人工智能将是类似的东西，造成很多颠覆，但可衡量的经济增长不多，而在2010年代研究这个问题的经济学家呢，往往得出的估计是，互联网和呢社交媒体免费服务的所有好处基本上都流向了消费者，但是呢，但是但是并没有以体现在你们可衡量的经济产出中的方式。

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🎧 I think with AI, you know, it's it's it's possible to imagine something similar uh happening.

我认为对于人工智能，你知道，有可能想象类似的事情发生。

🎧 You know, if AI makes internet search much better, it's a similar sort of thing.

你知道，如果人工智能让互联网搜索变得好得多，那是类似的事情。

🎧 Um if it makes just the access of uh knowledge easier, then again, it's a similar sort of thing.

嗯，如果它只是让获取知识更容易，那么 again，也是类似的事情。

🎧 The key question here as it is for the explosive growth scenario more generally is does AI help humanity really push forward the frontier of knowledge?

这里的关键问题，就像对更普遍的爆炸性增长情景一样，是人工智能是否真的帮助人类推动知识前沿？

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🎵 Um and if it does then I would say that probably puts AI in a different bucket.

嗯，如果它确实如此，那么我想说这可能将人工智能归入不同的类别。

🎵 You know, if it can uh be creative, come up with research ideas, accelerate the pace of transformation, then you have a potentially quite powerful impact on on longr run living standards uh from from that in a way that perhaps the internet, at least in observer in an observable way, didn't bring about.

你知道，如果它能呢具有创造性，提出研究想法，加速变革的步伐，那么它就有可能对长期生活水平产生相当强大的影响，呃，以一种或许互联网（至少以可观察的方式）未能带来的方式。

